

- (a)** Explain why it is important for plants to open and close their stomata in a daily rhythm.

..... [3]

-
- open stomata / %
- time / hours
- Key
- day
- night
- | Time (hours) | Open Stomata (%) | Period |
|--------------|------------------|--------|
| 0 | 15 | Day |
| 6 | 90 | Day |
| 12 | 75 | Day |
| 18 | 45 | Day |
| 24 | 25 | Night |
| 30 | 75 | Day |
| 36 | 90 | Day |
| 42 | 55 | Day |
| 48 | 35 | Night |
| 54 | 80 | Day |
| 60 | 70 | Day |
| 66 | 50 | Day |

Fig. 2.1

Fig. 2.1 shows that the percentage of open stomata increases in the first seven hours of the experiment. Describe the sequence of changes that occurs in the guard cells that leads to the stomata opening.

This image shows a full page of white paper with ten evenly spaced horizontal dashed lines, typical of primary school handwriting practice paper. The lines extend across the entire width of the page, leaving small gaps between them for writing. There are no margins, text, or other markings on the page.

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..... [6]

- (c) The experiment was repeated with *A. thaliana* plants that were left in darkness from 14 to 96 hours. The results are shown in Fig. 2.2.

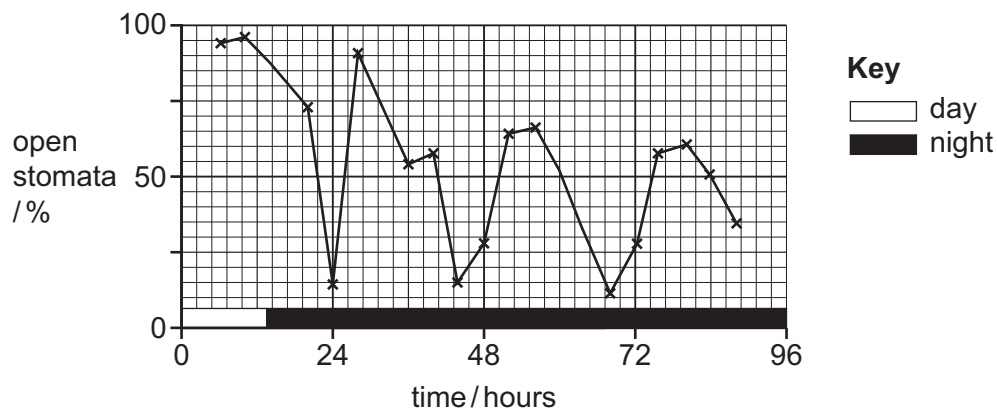


Fig. 2.2

With reference to Fig. 2.1, explain what Fig. 2.2 shows about the role of genes and the role of the environment in controlling the rhythm of stomatal opening and closing in *A. thaliana*.

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..... [2]

[Total: 11]